

Lecture 3: More infinity

Recall: $|\mathbb{N}| = |\mathbb{Z}|$ from last time

WTF??? Questions?

What is $|\mathbb{Q}|$?

Guesses?

Can we come up with a 1-1 Mapping Between $\mathbb{Q}$ and $\mathbb{Z}$?

Recall that $|\mathbb{N}| = |\mathbb{Z}|$, also Recall $\mathbb{Z} \subseteq \mathbb{Q}$

$$\dots -1, 0, 1, \dots = \dots -\frac{1}{1}, \frac{0}{1}, \frac{1}{1}, \dots$$

So we can try a few ideas:

Mapping $\mathbb{Z}$ to $\mathbb{Z} \subseteq \mathbb{Q}$ (integers to integers in $\mathbb{Q}$)

↳ works, but covers no fractions!!

Map all numbers of the same factors together:

$$1 \rightarrow \frac{1}{1}, \frac{2}{1}, \dots$$

works, but isn't 1-1

$$\frac{1}{1} \rightarrow 1 \text{ But so does } \frac{2}{1} \text{ and } \frac{3}{1} \dots$$

Answer is:

first consider the mapping $f: \mathbb{N} \rightarrow \mathbb{Z}$

(from last time)

if we can show if $|\mathbb{Q}| \stackrel{?}{=} |\mathbb{Z}|$, By extension.

We can show $\|N\|_2^2 \leq \|A\|_2^2$. (this is actually quite hard)

try this:

Handwritten diagram illustrating the steps of the Euclidean algorithm for finding the GCD of 12 and 18:

$$\begin{array}{rcl}
 \frac{1}{1} & \frac{1}{2} & \rightarrow \frac{1}{3} \dots\dots \\
 \downarrow & \nearrow & \\
 \frac{2}{1} & \frac{2}{2} & \nwarrow \frac{2}{3} \dots\dots \\
 & & \\
 \cancel{\frac{3}{1}} & \cancel{\frac{3}{2}} & \\
 \downarrow & \rightarrow & \\
 \frac{4}{1} & 2 & \\
 \downarrow & & \\
 \frac{1}{1} & & \\
 \vdots & & \\
 \vdots & & \\
 \vdots & &
 \end{array}$$

"Diagonalization"

$$\pi(x, y) = \frac{1}{2} (x+y)(x+y+1) + y$$

illustrating the fact that $|N| \neq |Q|$ is hard, But it's cool!

So Now ...

$$|\mathbb{N}| = |\mathbb{Z}| = |\mathbb{Q}|$$

Lastly what about $\mathbb{R}$?

it might make sense that since $\mathbb{Z} \subseteq \mathbb{Q}$ and $|\mathbb{Q}| = |\mathbb{Z}|$, it might make sense that $|\mathbb{R}| = |\mathbb{Q}|$.

However that is not the case.

Cantor's diagonal argument:

if we can show all numbers between 0 and 1 are larger than $|\mathbb{Q}|$
it would suffice to show that $|\mathbb{Q}| < |\mathbb{R}|$.

Say we can write out each number in $[0, 1]$
with one from $\mathbb{N}$.

0: .000...1
1: .123...345...9
2: .9134...
⋮
⋮
⋮

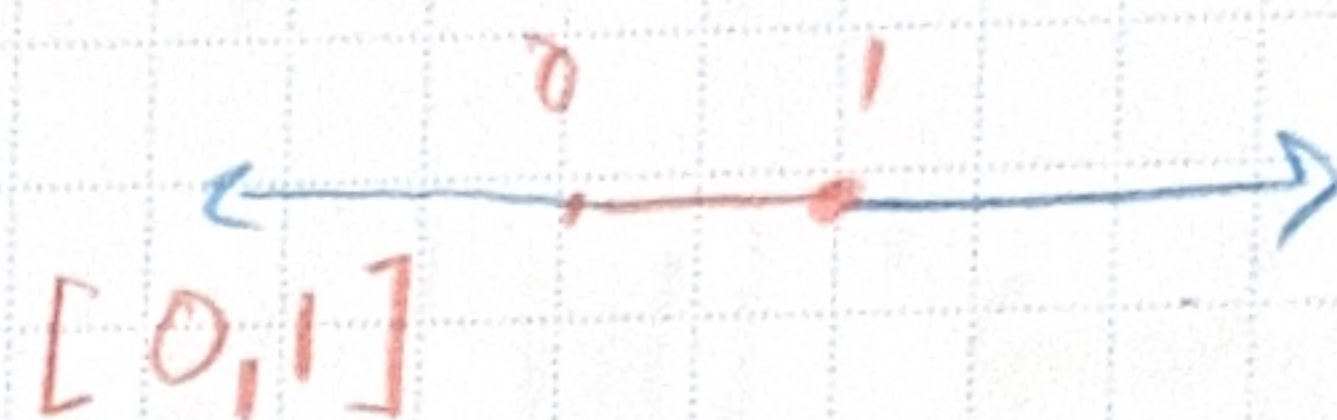
← like so

diagonal

Now for each digit in each number either increment by 1, or make it 0 if it's 9, then add the number to the list

to illustrate:

~~.123~~
~~.234~~
~~.345~~
3.246



Cantor's diagonal argument works because it ensures each number in the list is different in at least 1 place, thus ensuring its uniqueness.

We can repeat this forever, therefore no such list can exist. Therefore:

$$|[0,1]| > |\mathbb{N}| \rightarrow |\mathbb{R}| > |\mathbb{N}|$$

We call the size of infinity of $|\mathbb{N}|$ 'countable' as we can list all its elements given infinite time. This is aleph null $\aleph_0$.

The size of $|\mathbb{R}|$ is 'uncountable', in fact it has a special symbol. $|\mathbb{R}| = \aleph_1$, "aleph one".

One can frame the fundamental differences between $|\mathbb{R}|$ and $|\mathbb{N}|$ by looking for a "successor function."

on $\mathbb{N}$, the successor function, $S(n)$, answers the question what is the smallest larger number than n ?

on $\mathbb{N}$, the answer is simple. $S(n) = n+1$.

What about on $\mathbb{R}$?

try to find the next real number on $\mathbb{R}$ after 0!!!

$\frac{1}{2}$? no, $\frac{1}{4}$ is smaller. $\frac{1}{4}$? no, $\frac{1}{8}$ is smaller, and so on

the problem is, if you try to define a smallest next number on $\mathbb{R}$ you can always find a smaller one by, say, dividing by 2.

thus no successor on $\mathbb{R}$ exists.

~~—x—~~
 $\mathbb{R}$ and $\mathbb{N}$ feel similar, but in a way, $\mathbb{R}$ feels

"slippery" in my mind, there is no "last"

element in a list in $\mathbb{R}$.

More on this later...